

Dong Su Kim

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WORK EXPERIENCE

Conboy Lab

Machine Learning Research Associate

- Designed a **scalable distributed data pipeline** for multi-terabyte datasets, improving preprocessing speed 30% through parallelization, optimized I/O, and batching.
- Developed end-to-end ML workflows including feature extraction, model training, evaluation, and automated retraining hooks to maintain **98% uptime** for inference pipelines.
- Productionized ML pipelines using **Docker, CI/CD, and experiment versioning**, reducing configuration failures by ~25% across cloud and HPC environments.
- Implemented **model-serving infrastructure** with FastAPI and AWS (S3, EC2) to support async inference and multi-stage preprocessing.

PROJECTS

Sequence Modeling & ML Inference Pipeline

End-to-end system for protein sequence inference.

- Designed and deployed a **Dockerized ML** inference service integrating BERT/ESM2 embeddings with async **FastAPI** endpoints and cloud-backed storage.
- Built a **microservice pipeline** with preprocessing, embedding generation, model inference, and results aggregation supporting over ~50,000 sequences per run.
- Integrated **Postgres + S3** for structured metadata and bulk sequence storage, reduced data access latency by **35%**.
- Improved predictive performance **15%** through custom data augmentation and systematic hyperparameter tuning.

Hunger

Contextually Scalable Restaurant Rating Tracker - Beli Clone

- Built a full-stack restaurant discovery app indexing **1,000+ restaurants** with geospatial filtering and **<150ms** query latency via optimized geocoding and caching.
- Designed an ML-guided recommendation engine reducing **500+ candidates to top 1–3 matches** using a questionnaire, heuristics, and XGBoost ranking across **20+ features**.
- Developed a synthetic persona generator producing **10,000+ user profiles** with weighted cuisine distributions to enable offline evaluation of recommendation quality.
- Engineered **15+ modular async FastAPI endpoints** with reusable components and client-side validation, documented in a detailed design spec for maintainability.

Automated Camera Trap Classifier

Wildlife Conservation Platform

- Built an end-to-end CV training pipeline on a **100GB+(243,000 image) camera trap dataset** in AWS S3, restructuring 21 species into 6 conservation-relevant classes for modeling.
- Mitigated severe class imbalance by augmenting underrepresented classes, improved average class accuracy by **~12%**.
- Benchmarked baseline CNN (45% acc) vs. EfficientNet-B3 (**93% overall, 90% bobcat**), using results to guide model complexity and training strategy on AWS SageMaker.
- Developed an EDA and data quality pipeline to analyze blur, lighting, class imbalance, and annotation noise, directly informing augmentation and dataset redesign.

EDUCATION

UC Berkeley

Computer Science, Bioengineering

Berkeley, CA

SKILLS

Programming: Python, SQL, Java, C, JavaScript/TypeScript

ML & DL: PyTorch, TensorFlow, Keras, Hugging Face

ML Engineering: Model deployment, experiment tracking, distributed data processing, evaluation pipelines, GPU optimization

Data & Infra: Postgres, FastAPI, Docker, AWS (EC2, S3), CI/CD, Linux, HPC